

PROGRAM: DEADLOCK DETECTION ALGORITHM

C PROGRAM :

```
#include <stdio.h>
```

```
int main() {
```

```
    int allocation[10][10], request[10][10];
```

```
    int available[10];
```

```
    int work[10];
```

```
    int finish[10];
```

```
    int n, m;
```

```
    int i, j, k;
```

```
    int found;
```

```
    printf("Enter Number of Processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter Number of Resource Types: ");
```

```
    scanf("%d", &m);
```

```
    printf("\nEnter Allocation Matrix:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        for (j = 0; j < m; j++) {
```

```
            scanf("%d", &allocation[i][j]);
```

```
        }
```

```
    }
```

```
    printf("\nEnter Request Matrix:\n");
```

```
    for (i = 0; i < n; i++) {
```

```
        for (j = 0; j < m; j++) {
```

```
            scanf("%d", &request[i][j]);
```

```
        }
```

```
    }
```

```
    printf("\nEnter Available Resources:\n");
```

```

for (i = 0; i < m; i++) {
    scanf("%d", &available[i]);
    work[i] = available[i];
}

for (i = 0; i < n; i++) {
    finish[i] = 0;
}

do {
    found = 0;
    for (i = 0; i < n; i++) {
        if (finish[i] == 0) {
            for (j = 0; j < m; j++) {
                if (request[i][j] > work[j]) {
                    break;
                }
            }
            if (j == m) {
                for (k = 0; k < m; k++)
                {
                    work[k] += allocation[i][k];
                }
                finish[i] = 1;
                found = 1;
            }
        }
    }
} while (found);

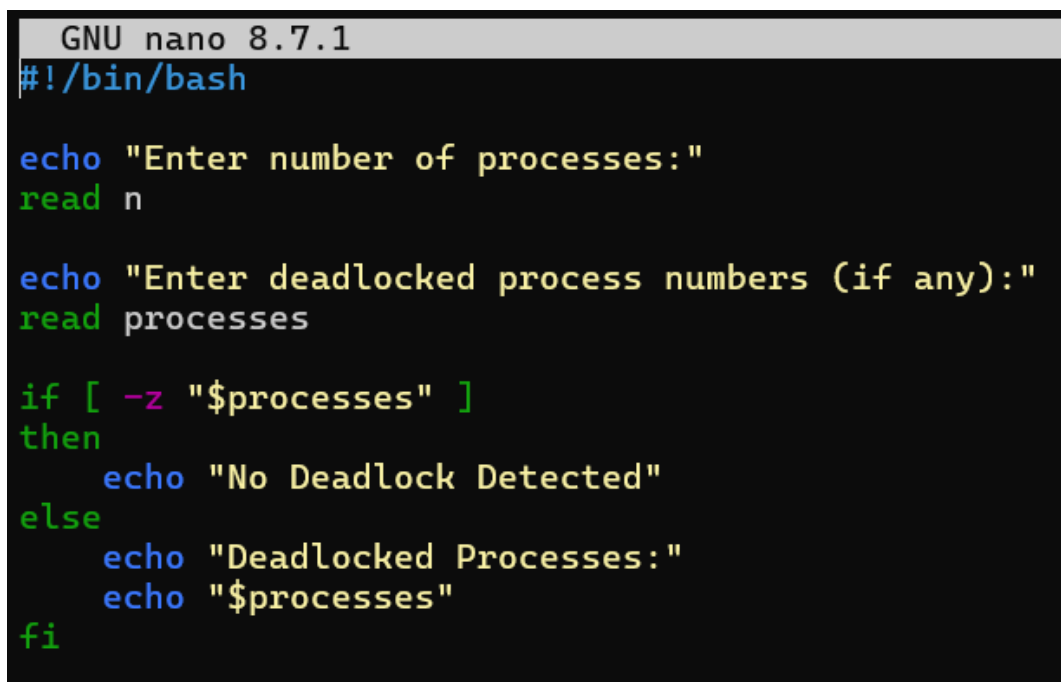
found = 0;

```

```

printf("\nDeadlocked Processes:\n");
for (i = 0; i < n; i++) {
    if (finish[i] == 0) {
        printf("P%d ", i);
        found = 1;
    }
}
if (found == 0) {
    printf("No Deadlock Detected");
}
printf("\n");
return 0;
}
SHELL SCRIPT

```



```

GNU nano 8.7.1
#!/bin/bash

echo "Enter number of processes:"
read n

echo "Enter deadlocked process numbers (if any):"
read processes

if [ -z "$processes" ]
then
    echo "No Deadlock Detected"
else
    echo "Deadlocked Processes:"
    echo "$processes"
fi

```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~  
$ nano bankers.sh
```

```
(ganesh@LAPTOP-R3RDH9NG)~  
$ nano deadlock.sh
```

```
(ganesh@LAPTOP-R3RDH9NG)~  
$ chmod +x deadlock.sh
```

```
(ganesh@LAPTOP-R3RDH9NG)~  
$ bash deadlock.sh
```

Enter number of processes:

3

Enter deadlocked process numbers (if any):

12

Deadlocked Processes:

12